

PRESENTATION SCRIPT

Our Final Dataset and Research Questions

Worth the Investment? Weekly Stakeholder Update
Data Retrieval and Wrangling

Thursday, September 3, 2026 · Presented by Mahmoud
Script follows the "Data_Retrieval_and_Wrangling.pptx" deck,
slide by slide

How to use this script

- Each card below matches one slide, in order. The gold number is the slide number, so it is easy to check you are on the right one.
- The main paragraph is what to say out loud. It is written in plain, spoken language, not slide text, so it will not sound like you are reading bullet points.
- The gold "if asked" boxes are optional. Only use them if a stakeholder actually asks a follow up question on that slide.

- You do not need to say every line word for word.
Treat this as a safety net, not a script to recite.

17

slides

~15

min talk time

9:00

prep, per email

10:00

discussion
starts

What this meeting needs to cover

Per the instructor's reminder: a walkthrough of the final dataset(s), and a walkthrough of the question set and research angles. No specific file format is required. Discussion starts at 10 am, following the order in the ideas sheet.

This deck covers both. Slides 1 to 12 build up to the final dataset (including how it was retrieved, combined, and explored, so the story of how you got there is honest and complete). Slides 13 and 14 are the research questions themselves, the second thing the instructor asked for directly.

OPENING LINE, BEFORE SLIDE 1

"Good morning, thank you for the time. Today I'm walking through our final dataset and our research questions, and along the way I'll show how we retrieved, combined, and explored the data to get there. This will take about fifteen minutes, and I'm happy to take questions any time."

THE SCRIPT · SLIDES 1 TO 17

1

TITLE SLIDE

Our Final Dataset and Research Questions

SAY

"This is our title slide for today. We're looking at the Worth the Investment project, player value and injury risk, and this is our weekly stakeholder update for September 3rd."

2

OUR BUSINESS QUESTION

Our Business Question

SAY

"Before anything else, here is the question our whole project is built around: **can we predict which football players are worth signing, based on how their market value might change and their risk of getting injured.** Everything I show today, the data, the process, the questions, all of it serves this one question."

3

PROCESS OVERVIEW

Before We Could Answer This

SAY

"To get to an answer, we go through six steps: ask, retrieve, combine, explore, clean, and analyze. Today's update covers steps one through five, how we shaped our questions, retrieved the data, combined it, explored it, and cleaned it. Analysis is next week's update."

What Shaped Our Data Retrieval

SAY

"Before we downloaded a single file, we had to decide what we actually needed. Three things shaped that. Our questions, like whether injuries affect a player's value. Our goals, we needed data that let us compare value, injuries, and performance for the same players. And only once those were clear could we judge whether a candidate dataset was actually useful, instead of collecting data at random."

The Three Datasets We Chose

SAY

"Based on that, we chose three datasets. Player transfer values and player injury records, both from Transfermarkt, sourced through Kaggle. And player performance stats, covering 2017 to 2024 across the top five leagues, from FBref, also through Kaggle."

IF ASKED WHY KAGGLE

Kaggle hosts cleaned, publicly shared versions of these datasets, so we didn't have to scrape the original sites ourselves. It's a normal, accepted source for a project like this.

SAY

"Getting from raw downloads to something usable took three steps. First, we downloaded the datasets as raw CSV files. Second, we organized them, keeping the raw files exactly as provided, tracked in Git so nothing gets lost or silently changed. Third, we loaded everything into a local database using dbt and DuckDB."

"We use DuckDB because it's a fast, free database that runs right on my laptop, no server to set up. dbt writes our cleaned tables into it, and we query it directly with plain SQL."

IF ASKED WHAT DBT IS

dbt is a tool that turns our data cleaning steps into small, named, reusable pieces of SQL, and runs them in the right order automatically, instead of one long manual script.

How We Combined the Data

SAY

"Three separate datasets don't help much sitting apart, so we combined them. Transfer values and injury records link together on the player's ID number. Performance stats link in on the player's normalized name, since that dataset doesn't share the same ID system. All three come together into one combined player dataset, built with dbt and stored in DuckDB."

IF ASKED WHAT "NORMALIZED NAME" MEANS

We standardized spelling, accents, and formatting of player names so that, for example, the same player isn't treated as two different people across datasets.

SAY

"Once combined, we didn't just trust the data, we checked it. We checked the date fields, which is actually what led us to find a real problem, I'll show that in a minute. We checked how many players matched across all three sources, which is what revealed a data gap. And we checked for duplicate records, to make sure each player only appears once."

SAY

"Exploring surfaced three real problems. First, two data tools on my computer were conflicting, so the wrong one kept running instead of the right one. We fixed that by always calling the correct tool directly, and the pipeline now runs cleanly every time.

Second, injury dates were stored as plain text instead of real calendar dates, which caused sorting mistakes. I'll show that fix on the next slide.

Third, about two thirds of players had no performance record after combining the datasets. We investigated that carefully, and I'll walk through it right after."

IF ASKED WHAT THE TOOL CONFLICT ACTUALLY WAS

My computer had two versions of a tool called dbt installed: dbt Cloud and dbt Core. This project only uses dbt Core, the free version that runs locally, but typing the short "dbt" command kept picking up the wrong one first, because of how the two were set up on my machine.

The fix was to call the correct one directly instead of the short command, so it always uses the right version now. I re-tested this again this week and it still runs clean, six out of six models, in under a second.

Deep Dive: Fixing Broken Dates

SAY

"Here's the date problem in more detail. Before, injury dates were plain text, so sorting them didn't actually work correctly. After, we converted every one into a real calendar date. The result: **99.98 percent** of all injury date records were fixed successfully, that's **107,951 out of 107,971** dates now reading correctly."

Deep Dive: A Missing Data Gap

SAY

"The second issue needed more care. **67 percent** of players, **1,310 out of 1,959**, had no matching performance record. We didn't just accept that number. We checked carefully to confirm it was a real gap in the data, not a mistake in our own process, and it was real. We also tested whether the missing players were meaningfully different, and on average they had a much lower market value. Rather than hide this, we're disclosing it clearly as a known limitation."

IF ASKED WHY THIS GAP EXISTS

FBref's performance data only covers the top five leagues from 2017 to 2024. Players outside that window, or in smaller leagues, simply don't have a performance record to match against, which lines up with why the missing players tend to be lower value.

SAY

"Putting it all together, here is where things stand. All six of our data steps run successfully, end to end. One local database. Three data sources combined into it. **1,959 players** in the final dataset. It's all built with Python, pandas, dbt, and DuckDB, and every step runs automatically, so if we add new data later, we can rebuild the whole dataset with a single command."

IF ASKED WHAT THE SINGLE COMMAND IS

It's called `dbt run`. In this project I actually type it as `"python -m dbt.cli.main run"`, for the same reason as the tool conflict fix on the earlier slide, that always calls the correct version. It re-runs all six of our data steps in the right order automatically, so I don't have to run each one by hand. One thing to be precise about if pushed: this command reshapes whatever data is already loaded into the database. If we added a genuinely new raw data source, we'd first re-run the download and load steps, that part isn't automatic yet, but once new data is loaded in, this one command rebuilds everything downstream by itself. I ran it again this week and it finished in under a second.

Our Question Set: Four Research Angles

SAY

"This is the second thing you asked me to cover today: our research angles. We have four questions, each written for a different audience.

For **model validation**: if we predict how a player's market value will move, is that forecast accurate enough to trust.

For **recruitment and scouting**: does how recently a player was injured predict whether their market value is about to fall.

For **sporting directors**: do younger players combine rising value with lower injury risk, compared to older players.

And for **medical and performance staff**: does a player's position on the pitch predict how much injury or workload risk they carry."

IF ASKED FOR THE ANSWERS ALREADY

These are the questions we're set up to answer. The actual statistical results are next week's update, data analysis and key findings, this week is about showing the data is ready and the questions are well defined.

How Our Dataset Answers Each Question

SAY

"Here's how our dataset actually supports each one. Value history powers the value forecast question and the sporting directors tradeoff question. Injury records, meaning dates and recency, power the recruitment and scouting question. Performance stats, specifically workload per ninety minutes, power the medical and performance position question, alongside injury records. And player profile, age and position, ties every question back to who the player actually is."

SAY

"This phase, data retrieval, wrangling, and research framing, is complete. We shaped clear questions and goals before collecting anything. We collected three real world data sources. We combined them into one clean dataset. We explored the data and solved every problem we found. We fixed the data errors in the injury dates. We investigated and disclosed a data gap honestly. And we defined four clear research questions for analysis."

Where This Fits in Our Project Roadmap

SAY

"Zooming out for a second, here's where this fits in the bigger picture. Kickoff and planning, done. Data retrieval and wrangling, done, that's what today covered. Data analysis and key findings is up next. Then dashboards and presentation, and finally the capstone presentation itself."

IF ASKED FOR AN EXACT COMPLETION PERCENTAGE

Keep it qualitative: "We're tracking well against the roadmap, on schedule to move into analysis next." This slide is intentionally a phase view, not a precise number, so there's no need to volunteer one.

Next Steps & Thank You

SAY

"Next week's update covers data analysis and key findings, that's where we'll bring in the actual statistical results for these four questions. Thank you for your time, I'm happy to take questions now."

MOVING INTO DISCUSSION

Per the instructor's email, discussion starts at 10 am, following the same order as the ideas sheet. After slide 17, pause, take a breath, and let the room lead. It's fine to say "let me check my notes" before answering something specific, that reads as careful, not unprepared.

If a question is about results (which hypothesis is confirmed, what the numbers show), it's fine to say plainly that analysis is next week's deliverable, today was about the data and the questions.

Quick facts, in case you're asked

A fast lookup, so you don't have to hold every number in your head.

1,959	players in the final combined dataset
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3	data sources: Transfermarkt values, Transfermarkt injuries, FBref performance
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6 / 6	dbt models run successfully, end to end
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99.98%	of injury dates fixed (107,951 of 107,971)
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67%	of players with no performance match (1,310 of 1,959), confirmed real, disclosed
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4	research questions, one per stakeholder audience
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player_id	join key for values and injuries; normalized name for performance
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Python, pandas, dbt, DuckDB	the tools behind the whole pipeline
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